# Warm-up

Alexander Golovnev

# Outline

Airlines Graph

Knight Transposition

Seven Bridges of Königsberg

# Airlines Graph

Consider a small country with five cities: $A, B, C, D, E$.

There are six flights:

$A - B, A - C, A - E,$

$B - D, C - D, C - E.$

Is there a direct flight from A to D?
With one stop?
With two stops?

# Airlines Graph

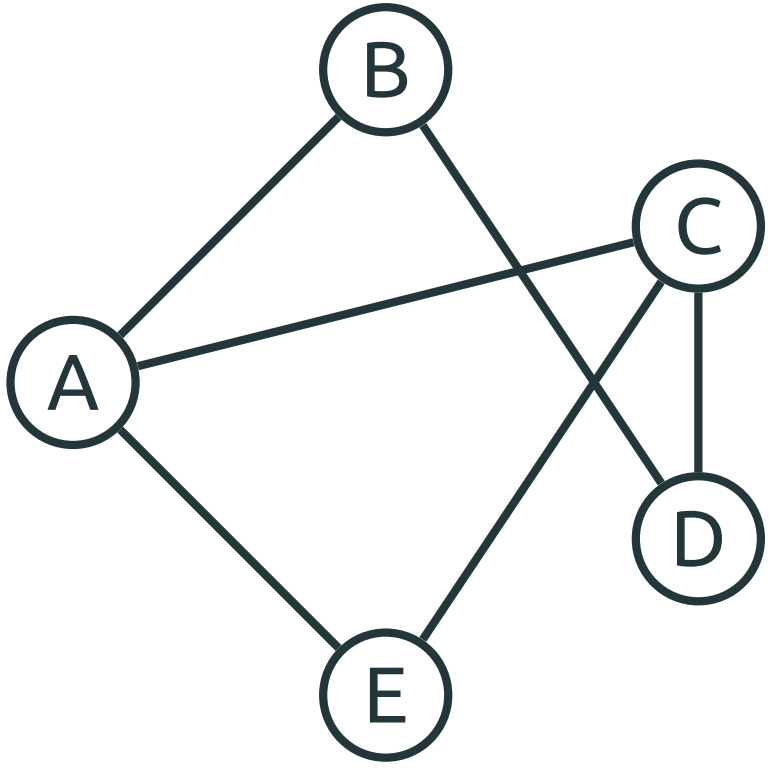
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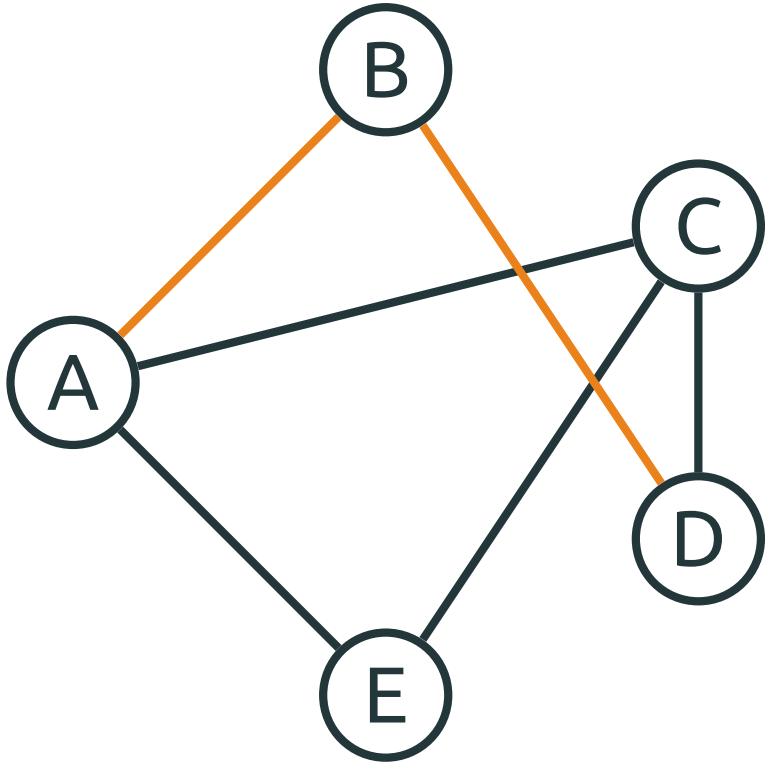
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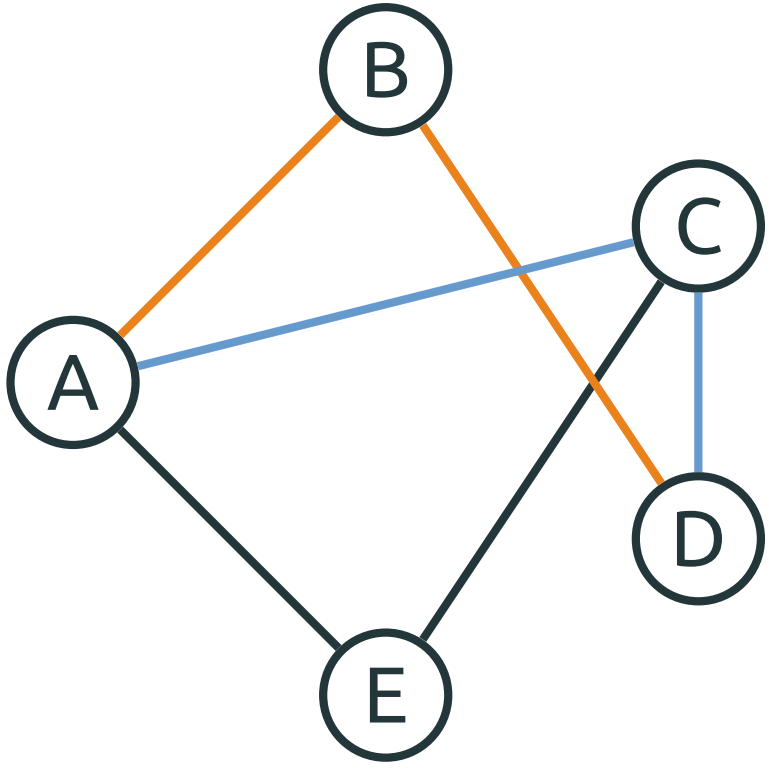
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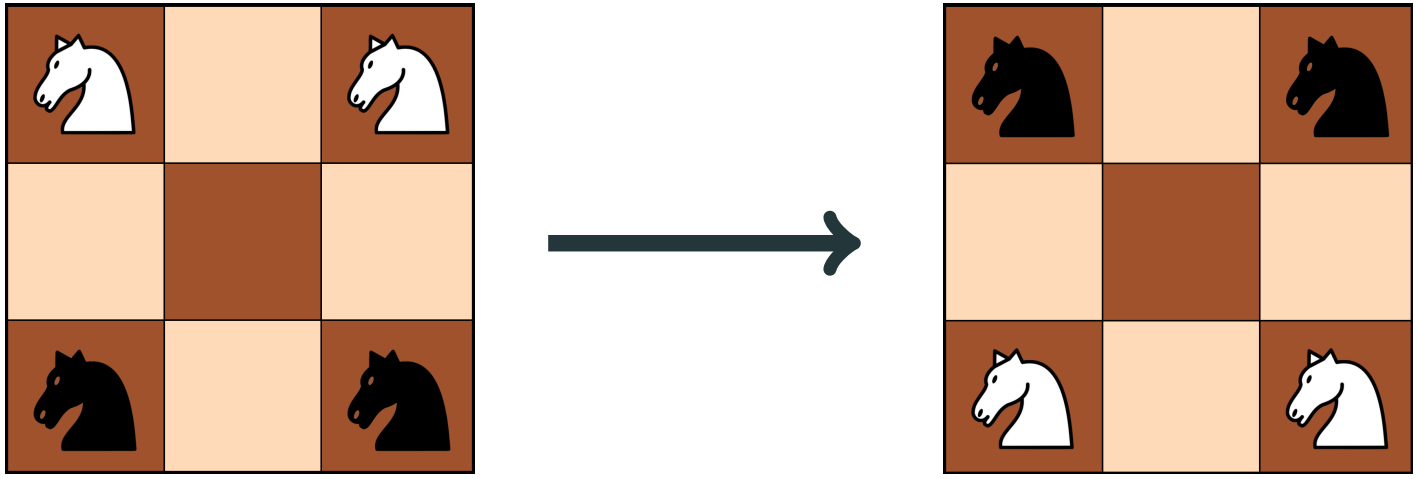
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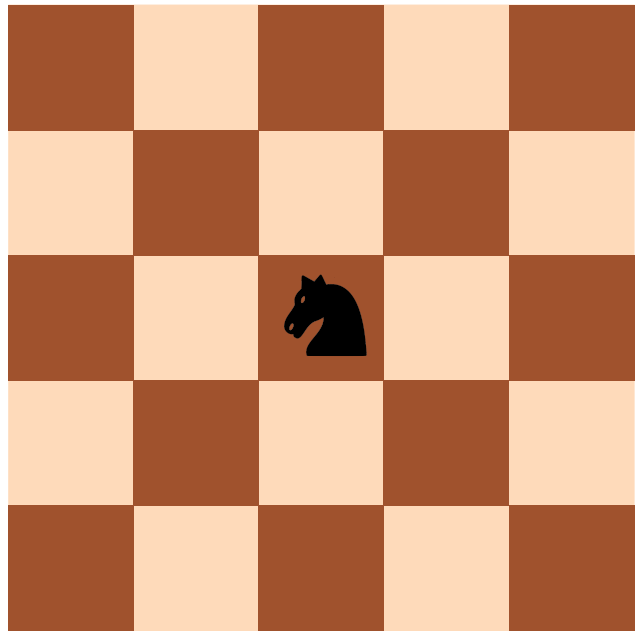
# Guarini's Puzzle



Exchange the places of the white knights and the black knights.

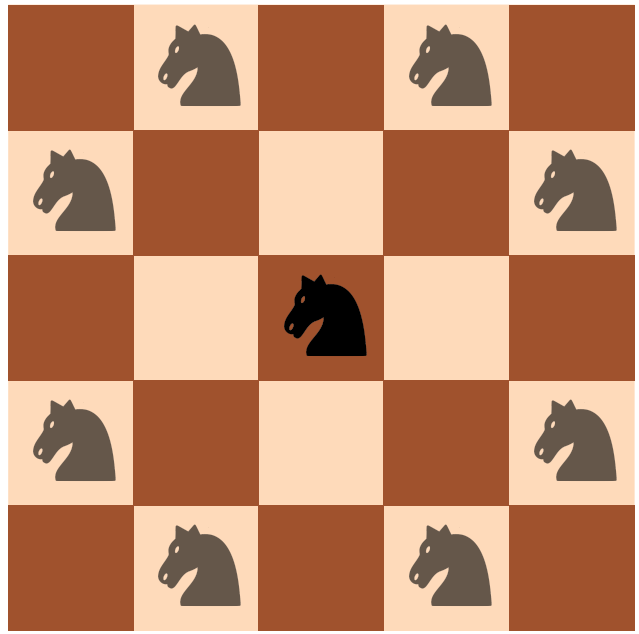
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A chess knight can move in an **L** shape in any direction

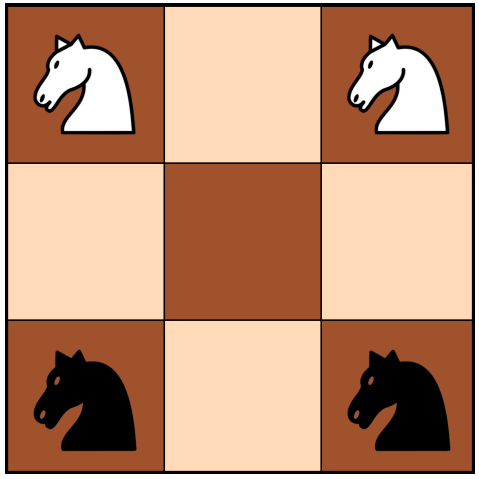


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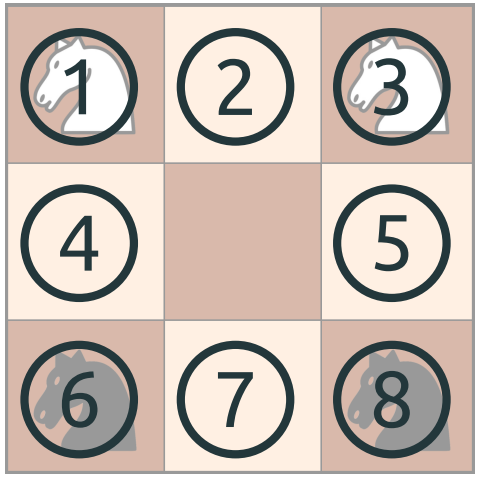


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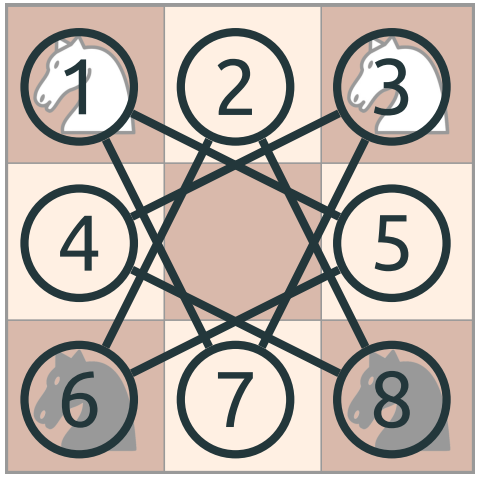
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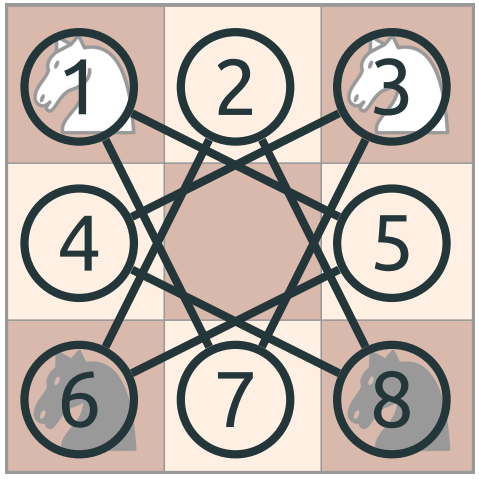

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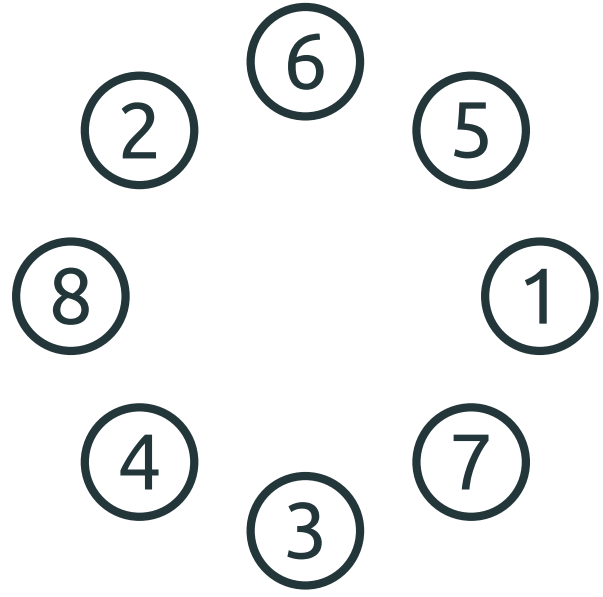
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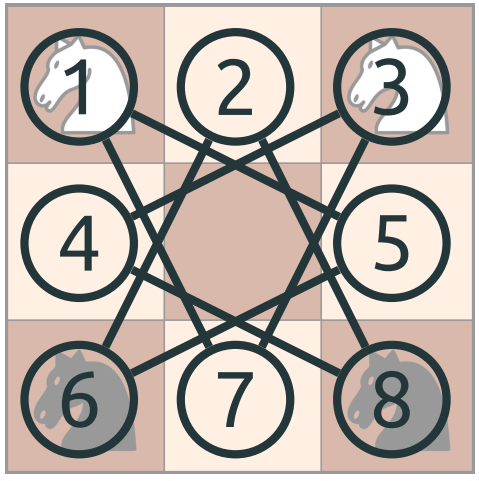
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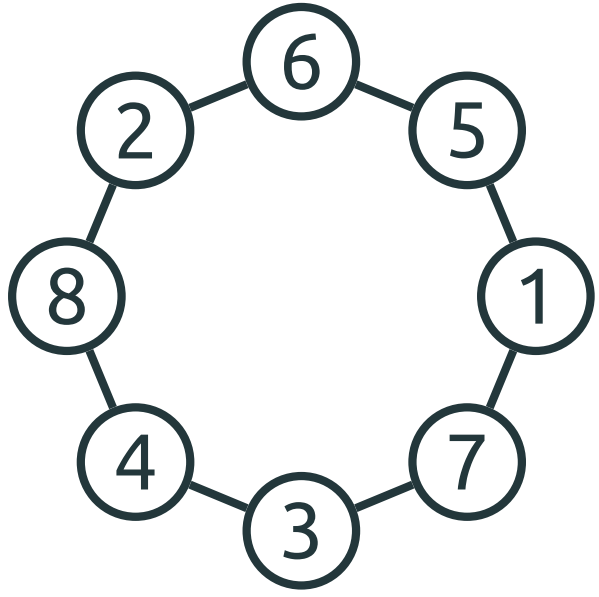



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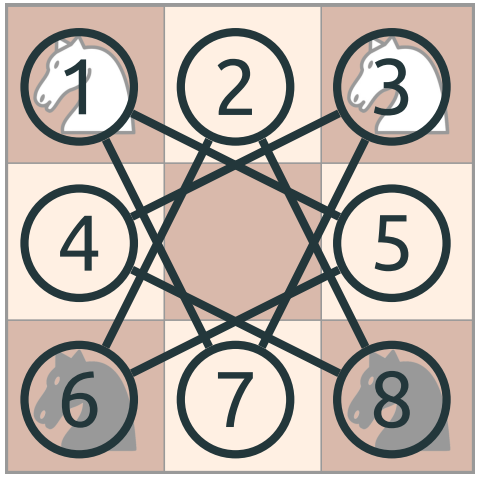
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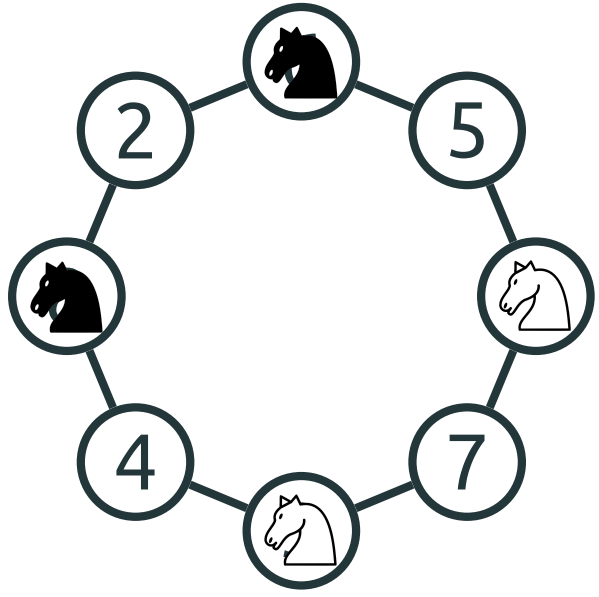



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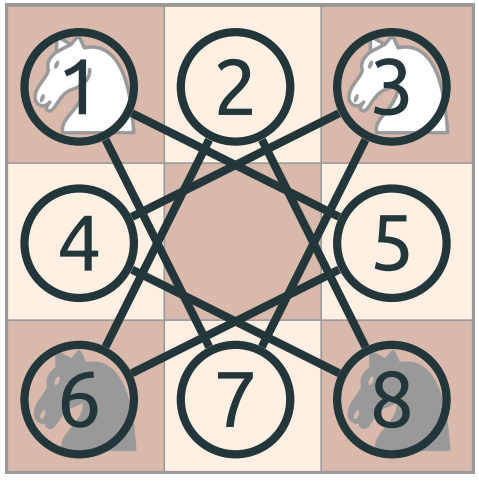
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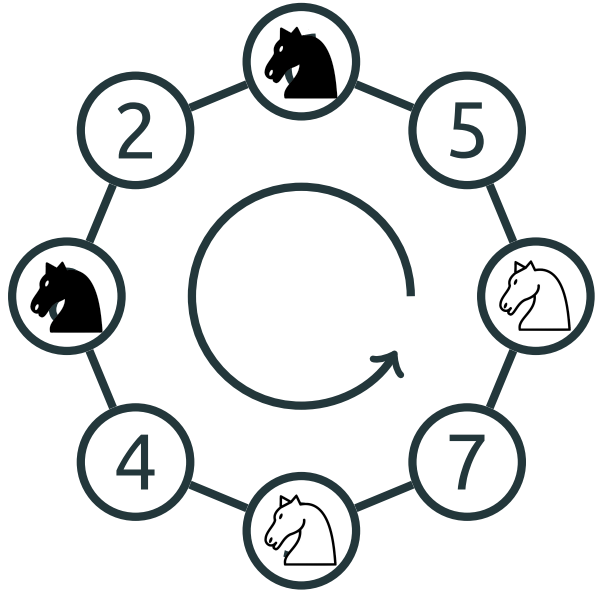



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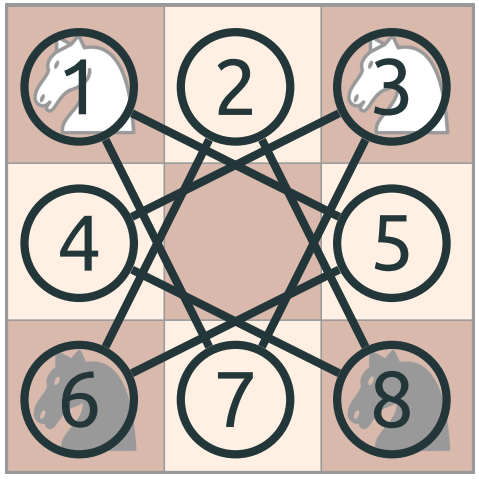
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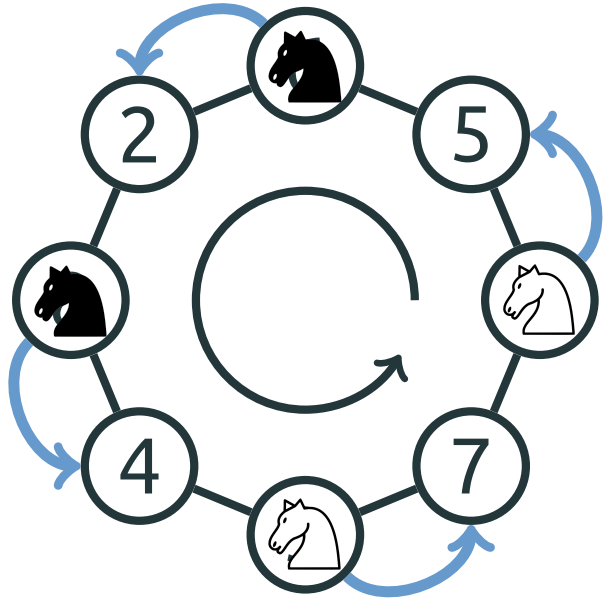



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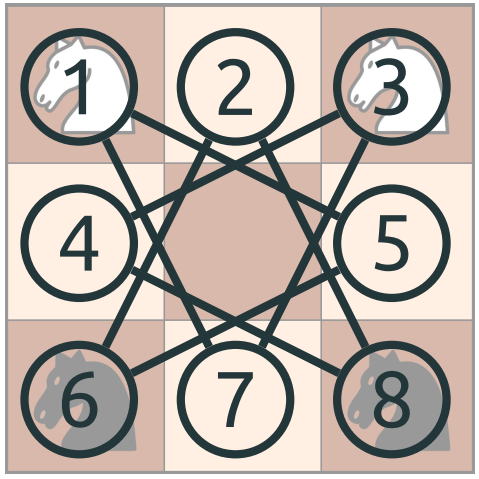
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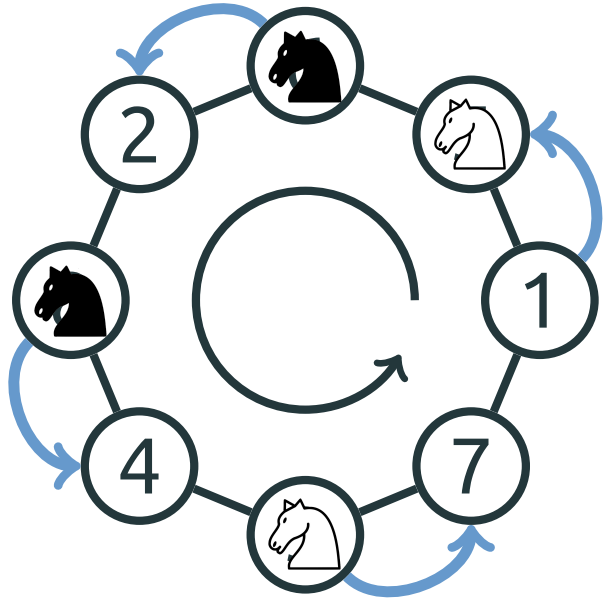



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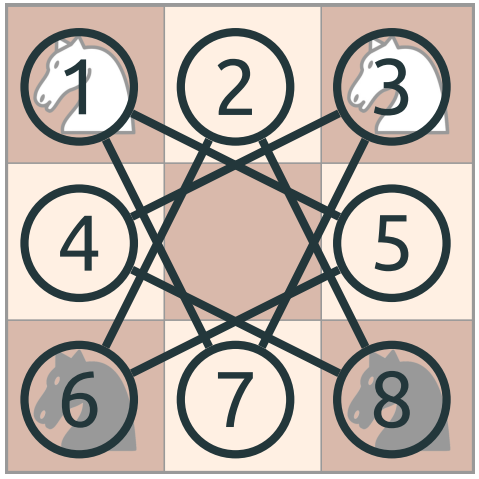
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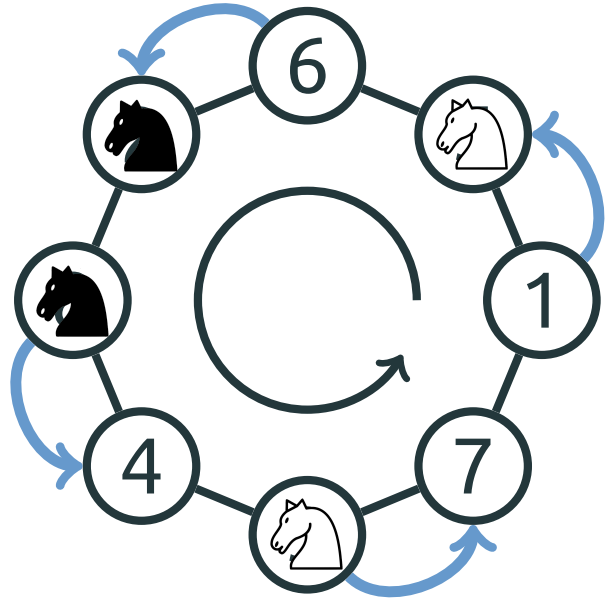



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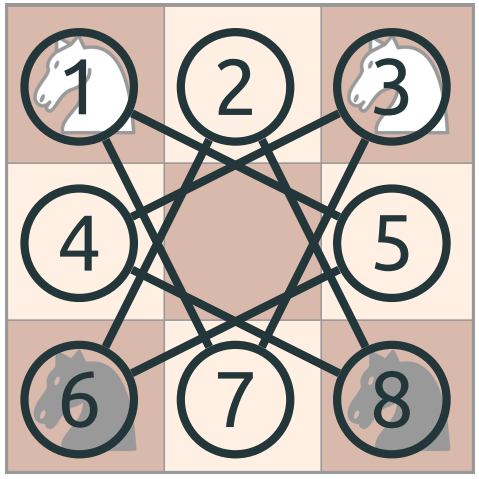
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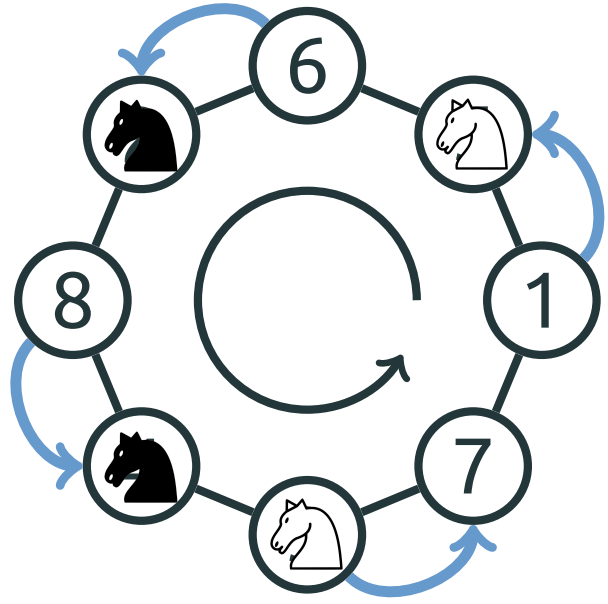



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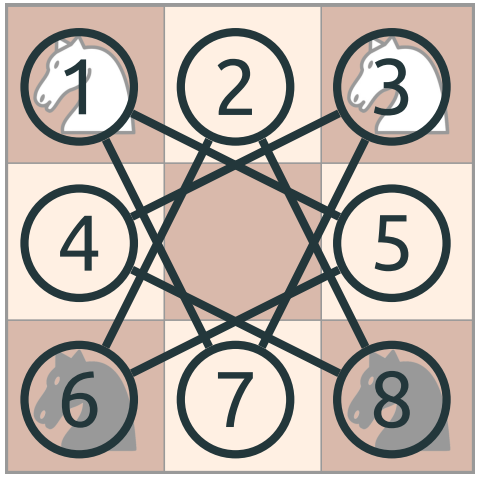
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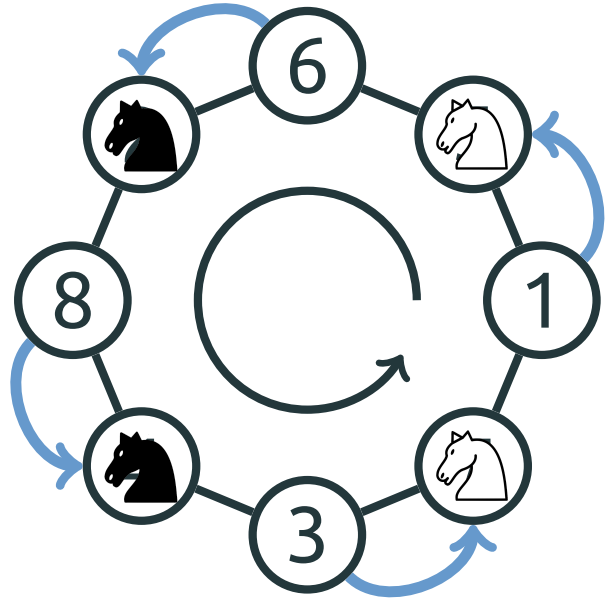



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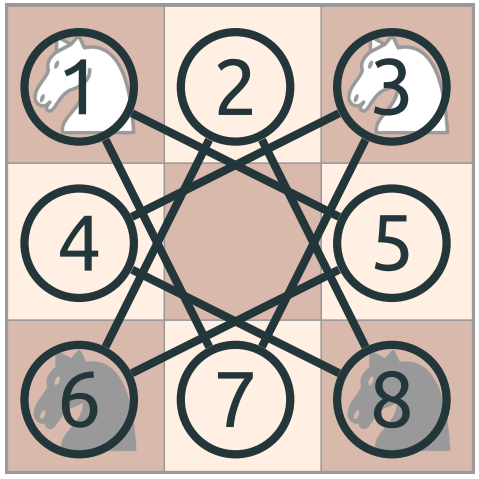
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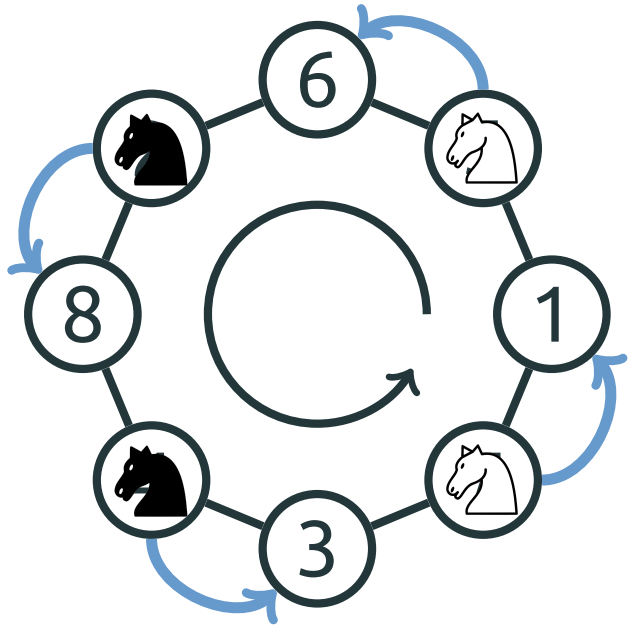



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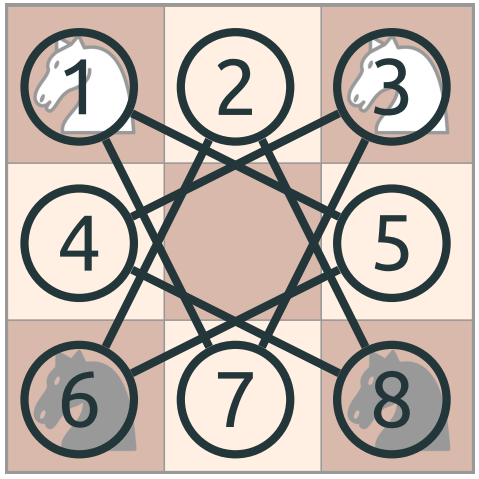
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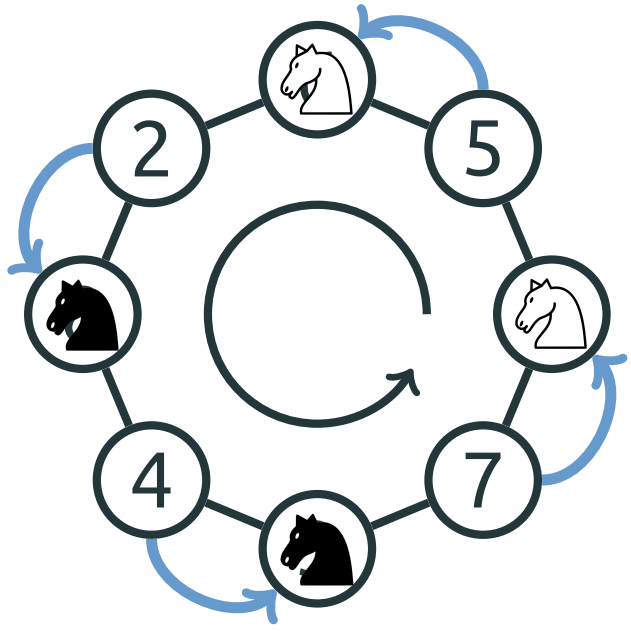



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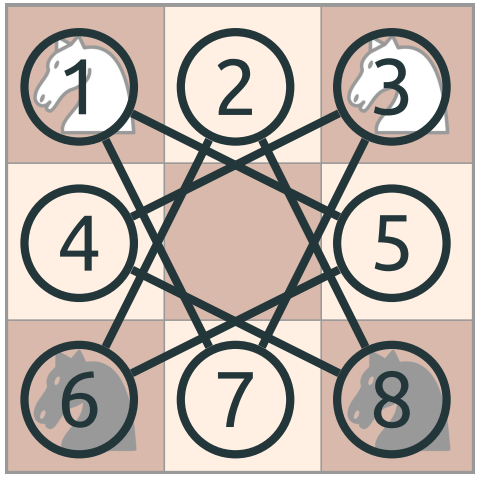
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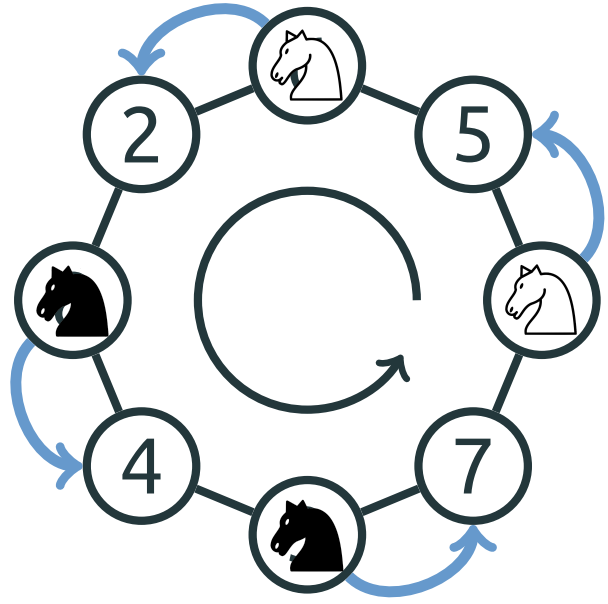



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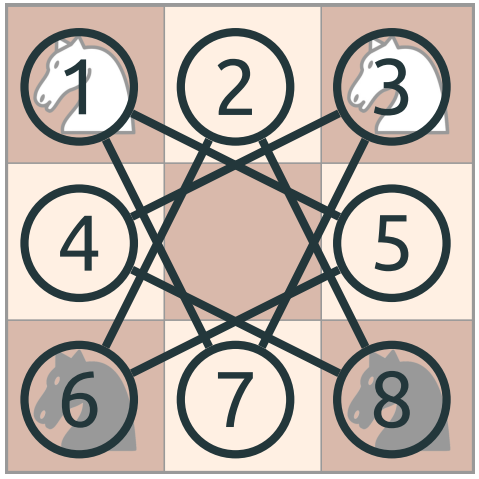
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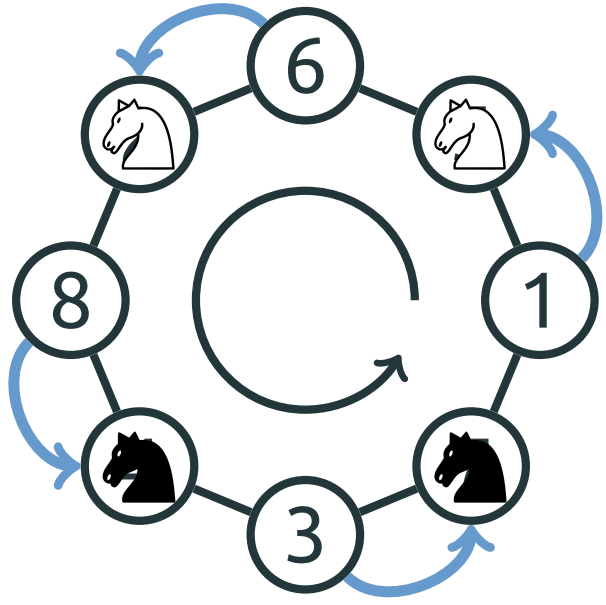



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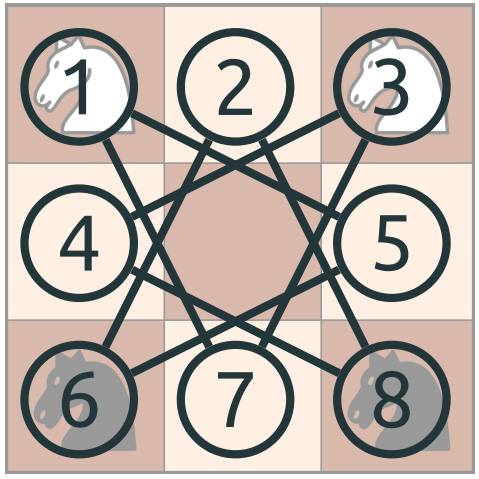
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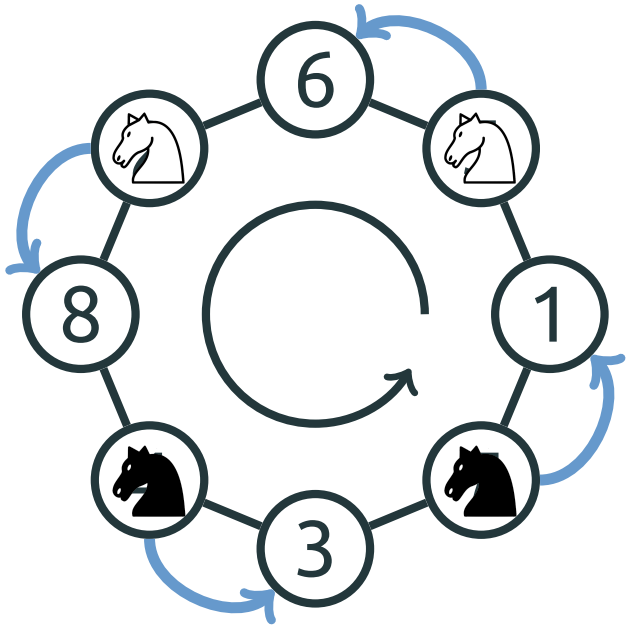



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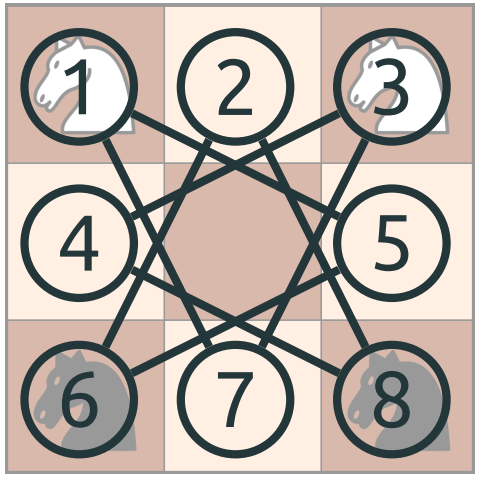
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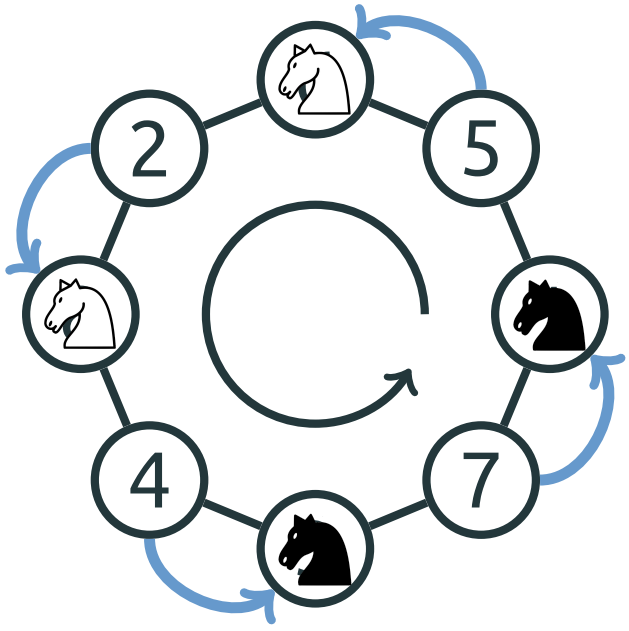



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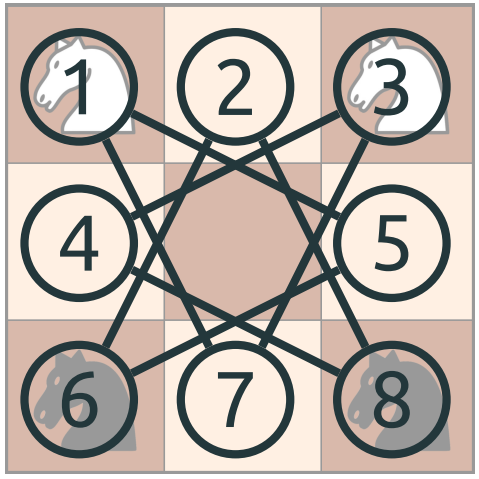
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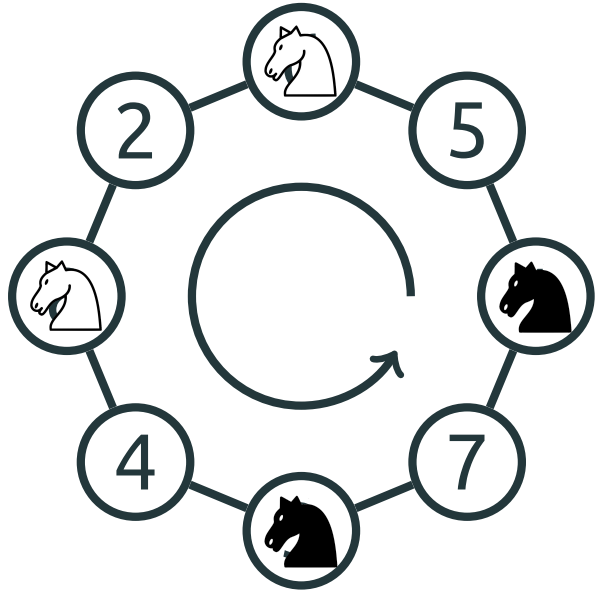



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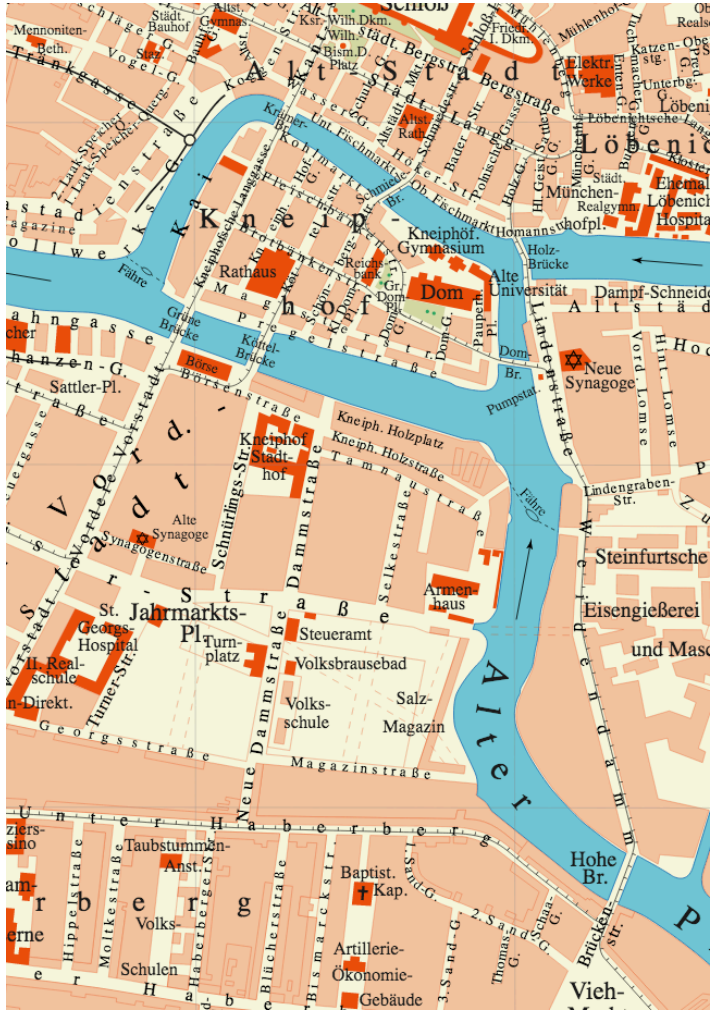
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# Seven Bridges of Königsberg

Königsberg, Prussia, 1735



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Walk through Königsberg

Cross each bridge
exactly once



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Leonhard Euler



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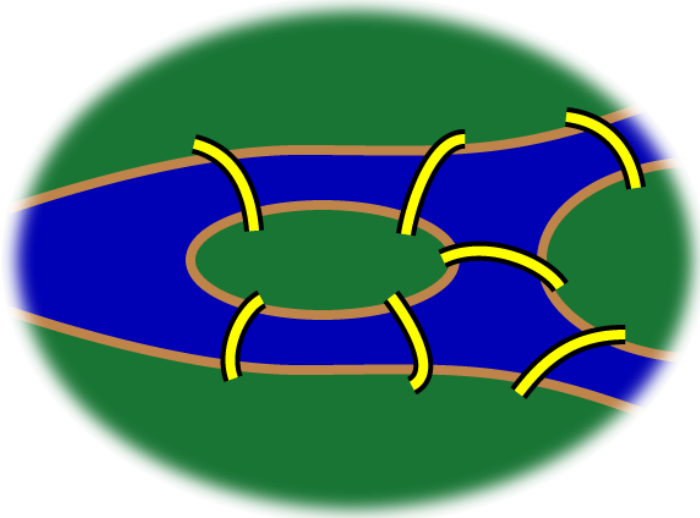


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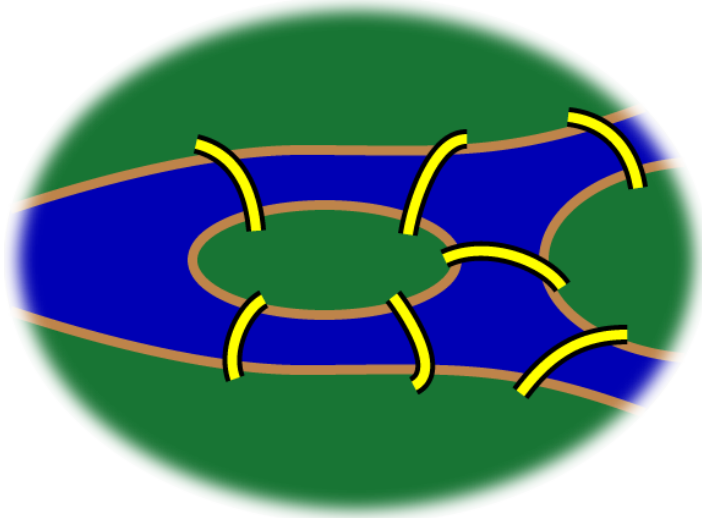


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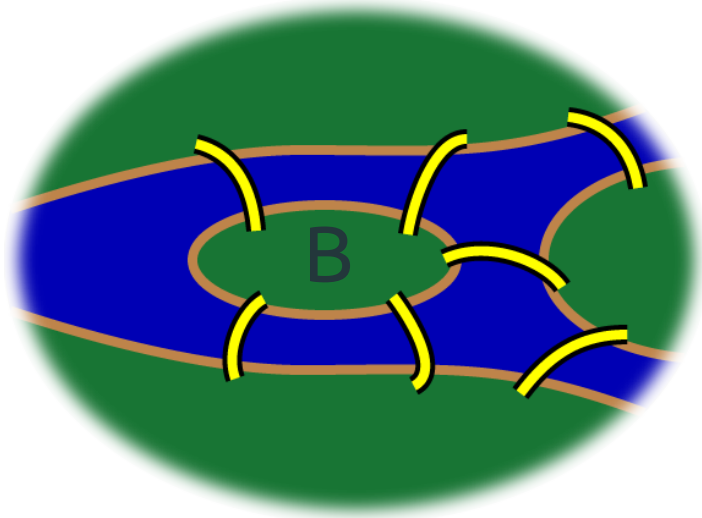


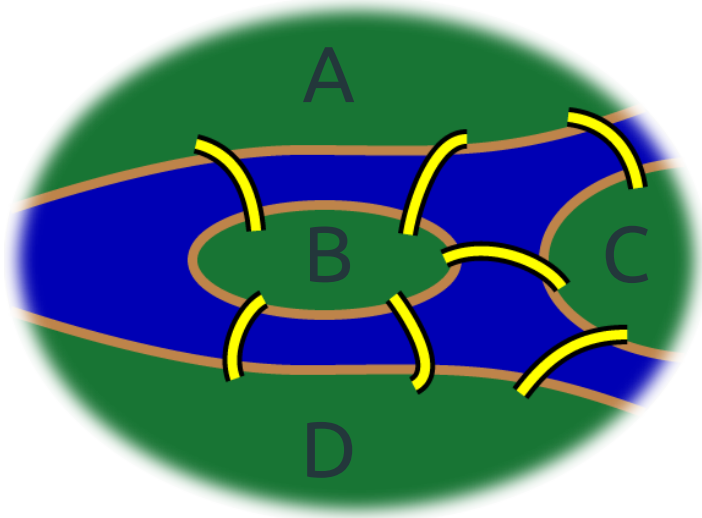
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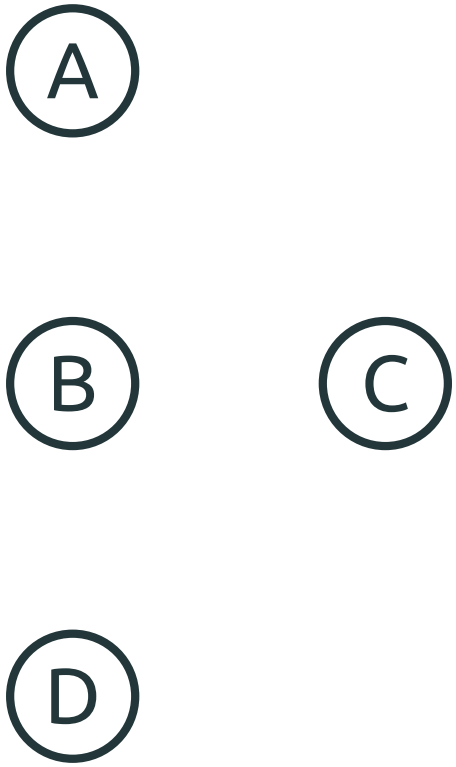
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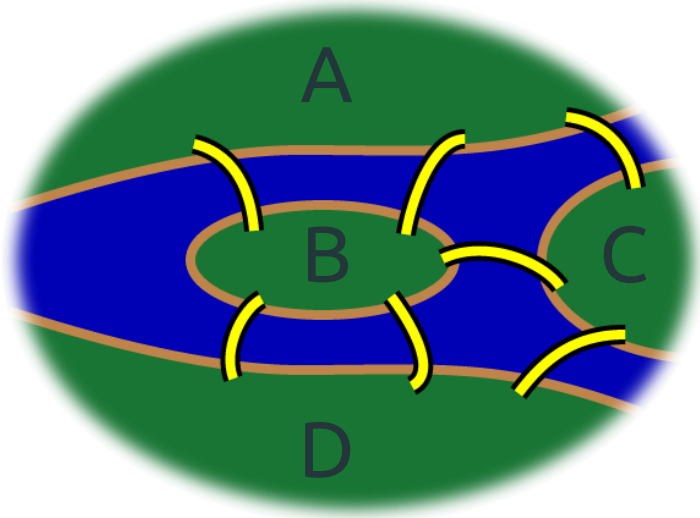


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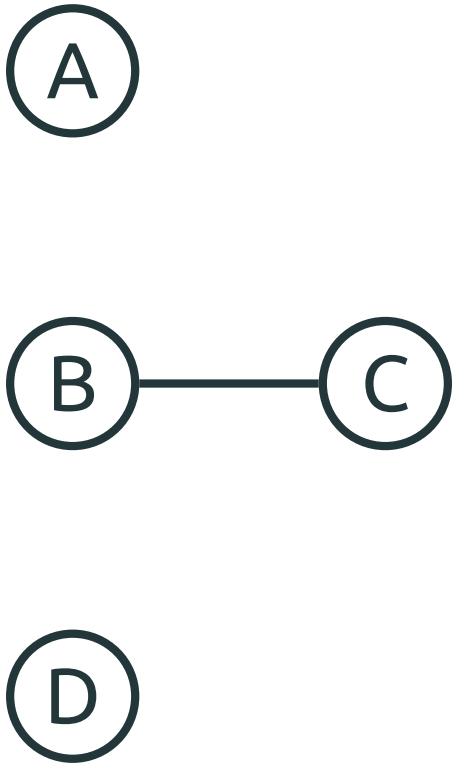


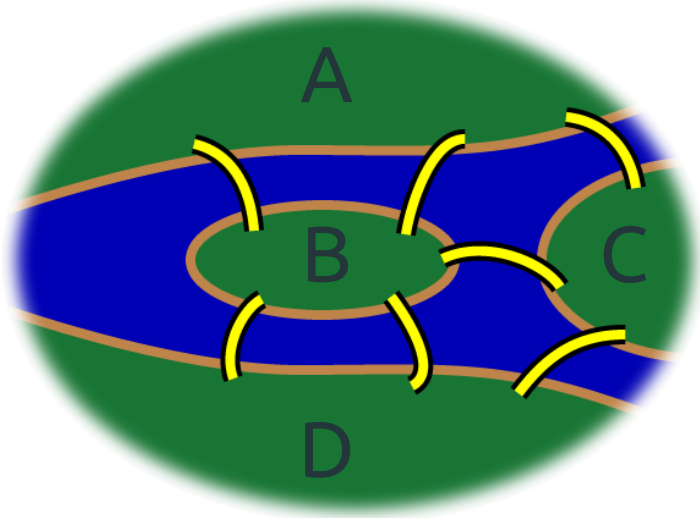
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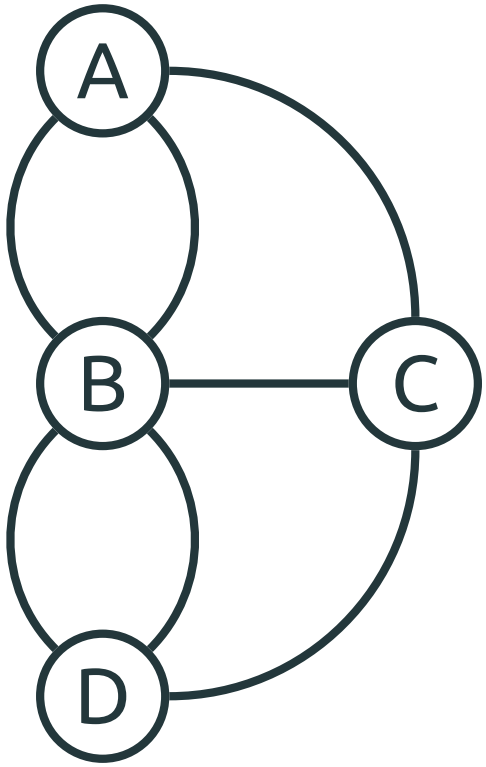



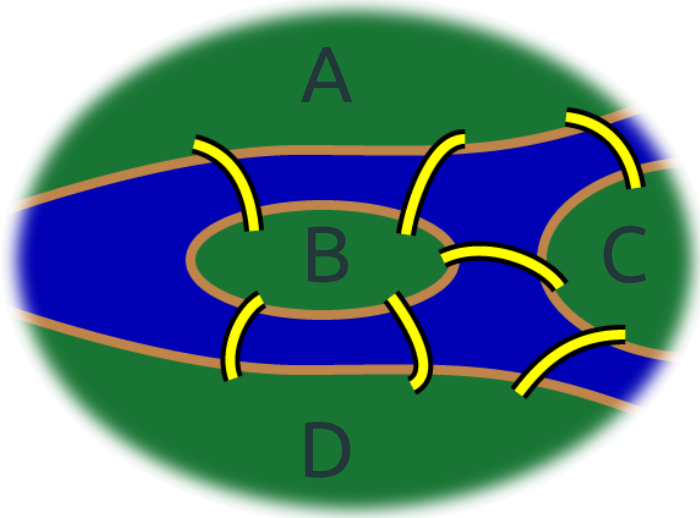
# Bridges of Königsberg. Graph



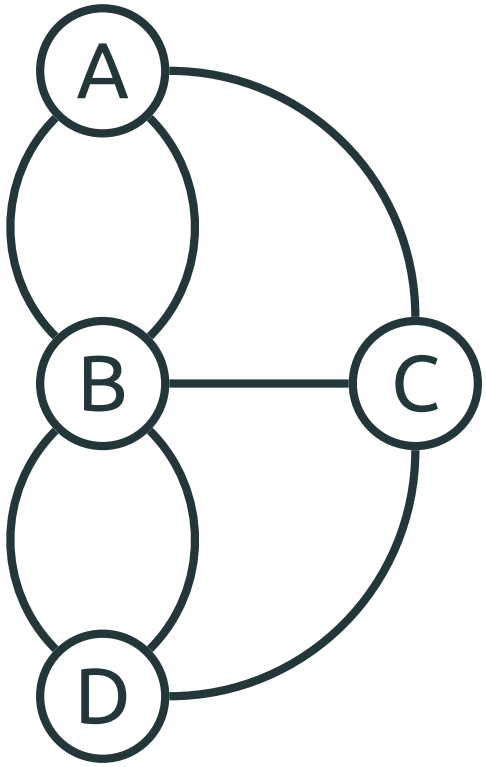



# Bridges of Königsberg. Graph



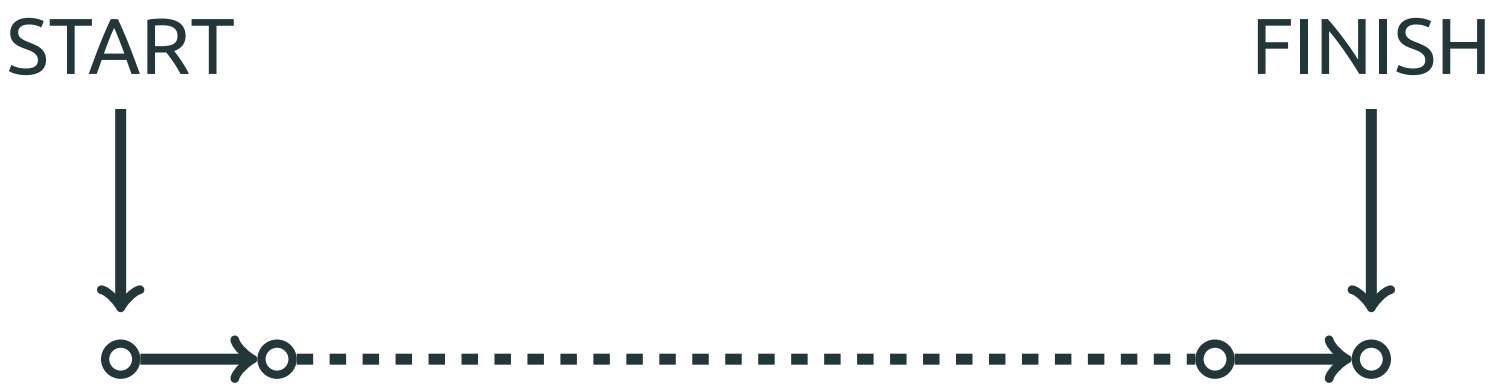
# Bridges of Königsberg. Graph



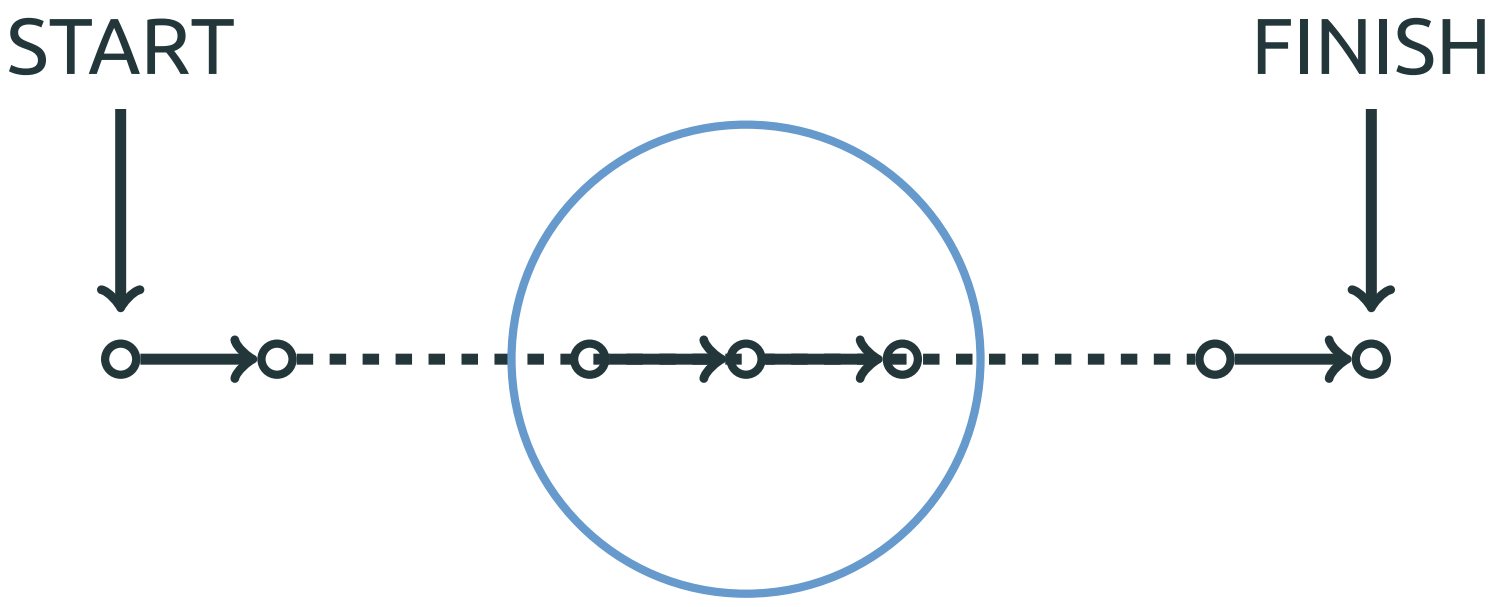

Is there a path which visits every edge exactly once?

Eulerian Path

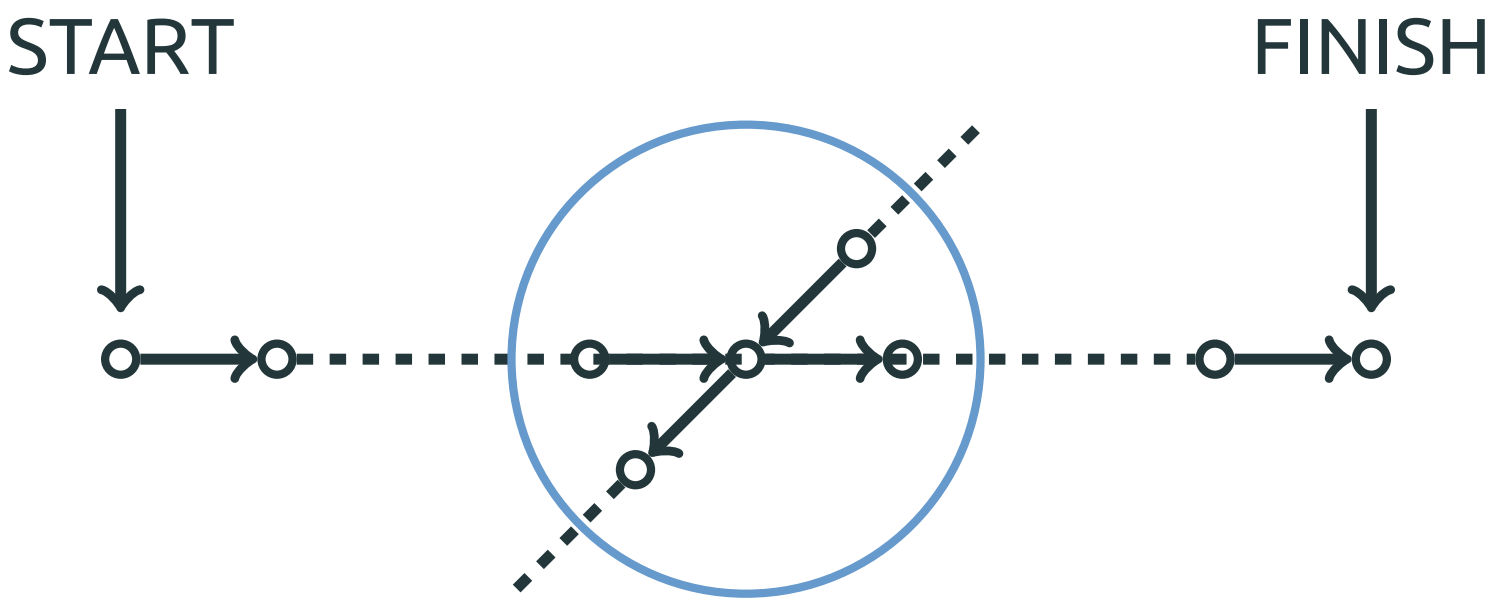
# Eulerian Path



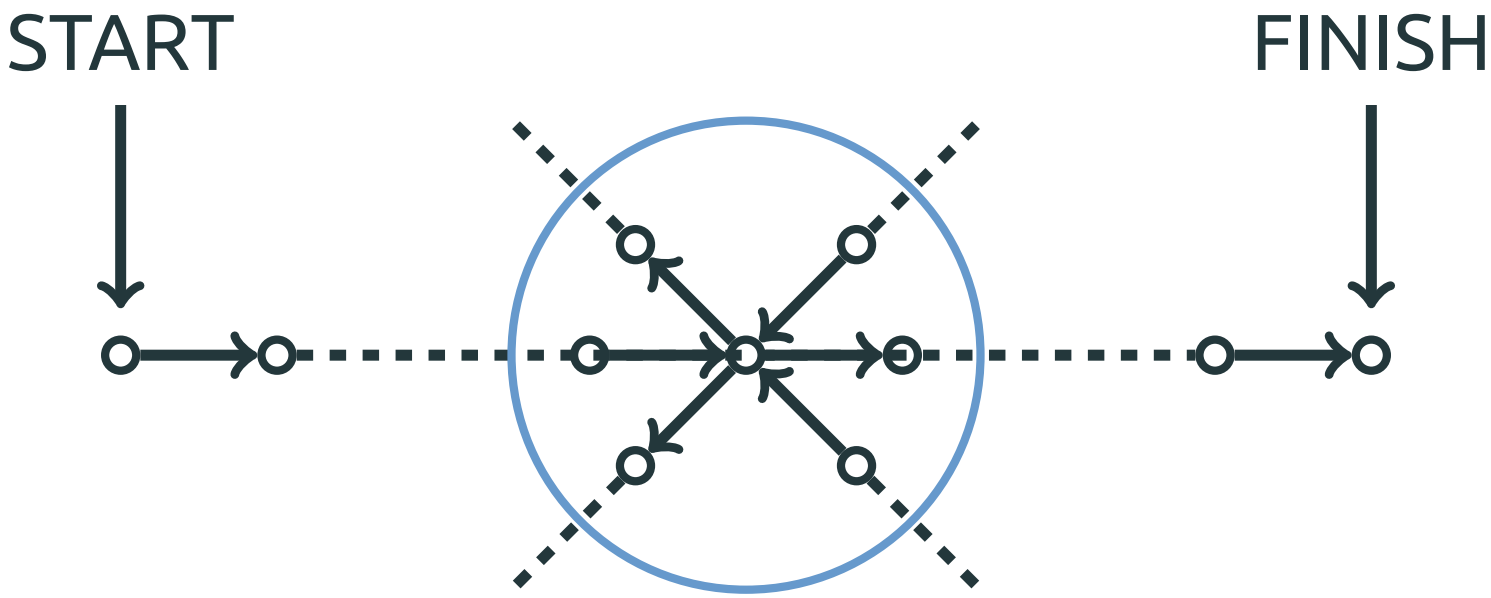
# Eulerian Path



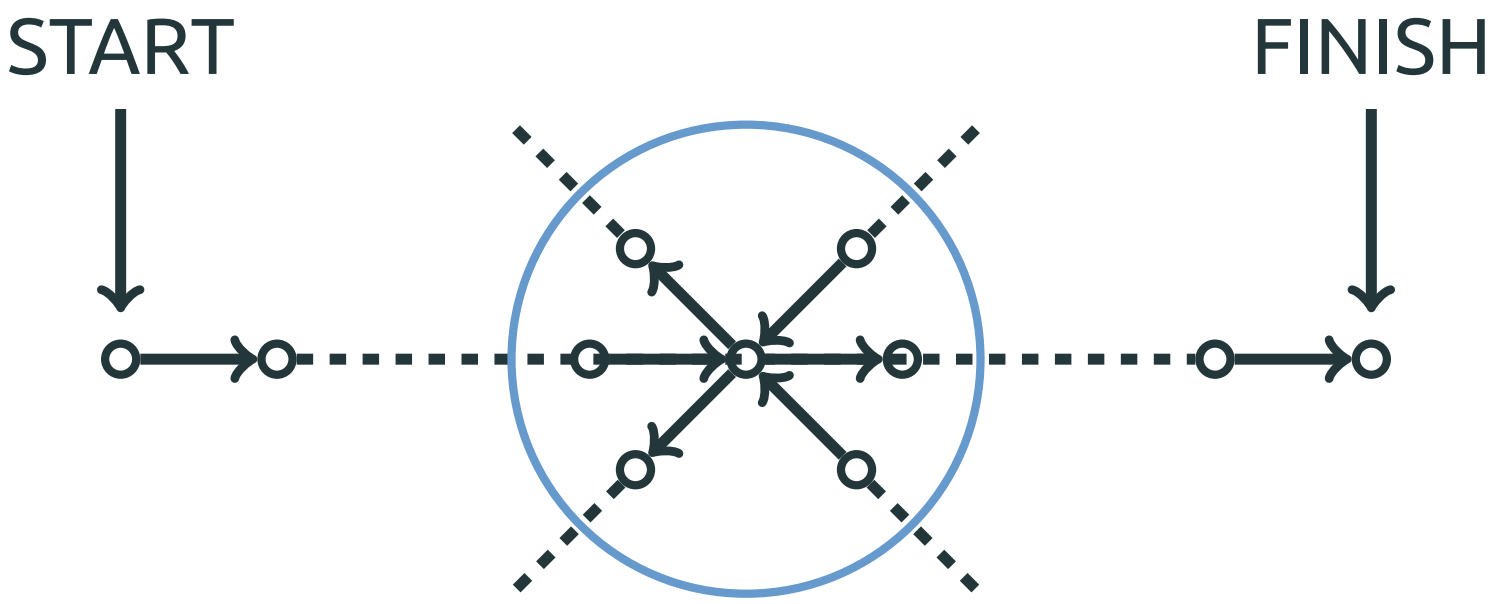
# Eulerian Path



# Eulerian Path

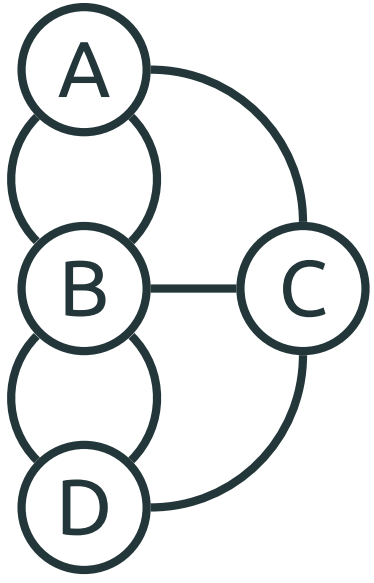


# Eulerian Path



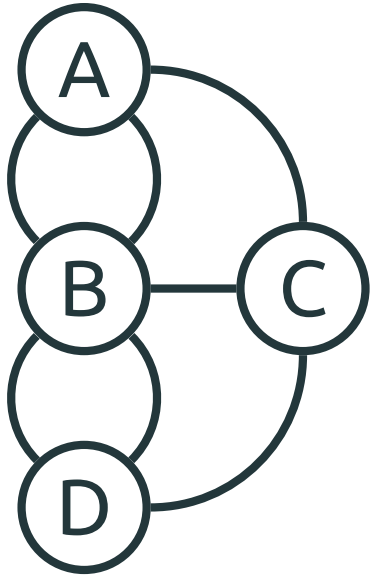

All but START and FINISH vertices
have even number of neighbors

# Eulerian Path




All but START and FINISH vertices have even number of neighbors

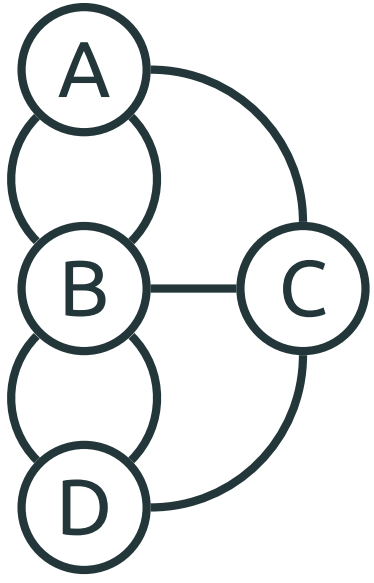
# Eulerian Path




All four vertices have
odd number of neighbors

All but START and FINISH vertices
have even number of neighbors

# Eulerian Path

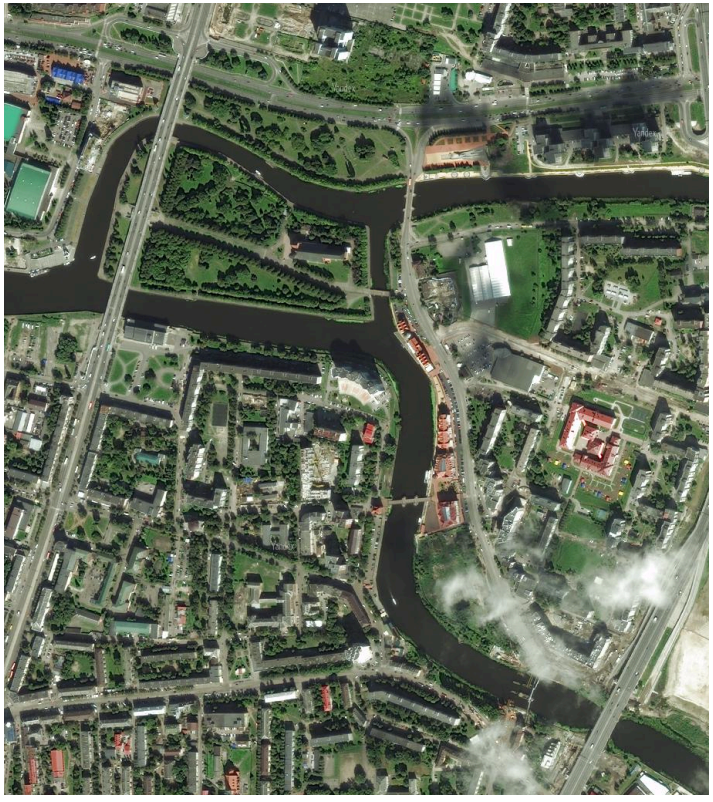



All four vertices have
odd number of neighbors
Impossible!

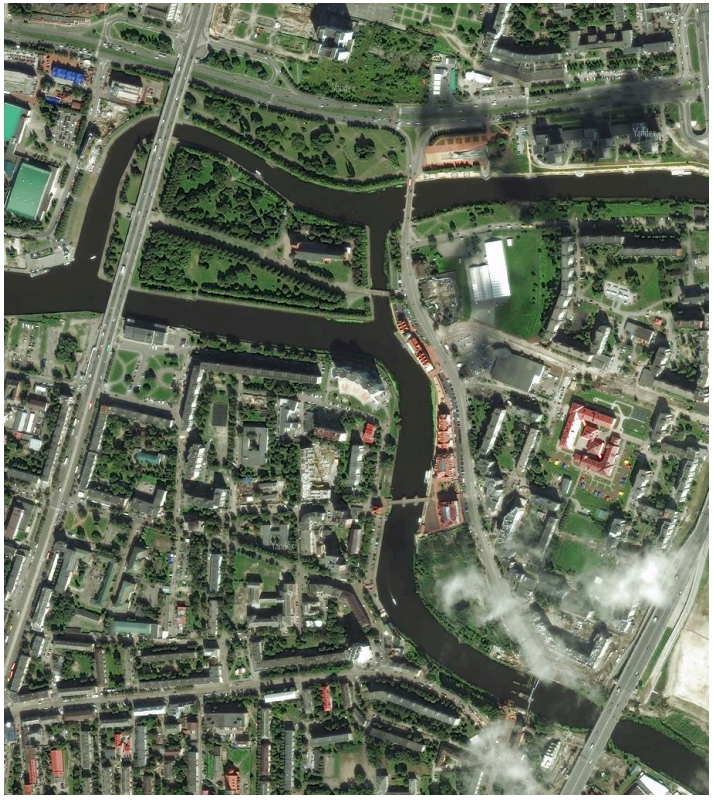


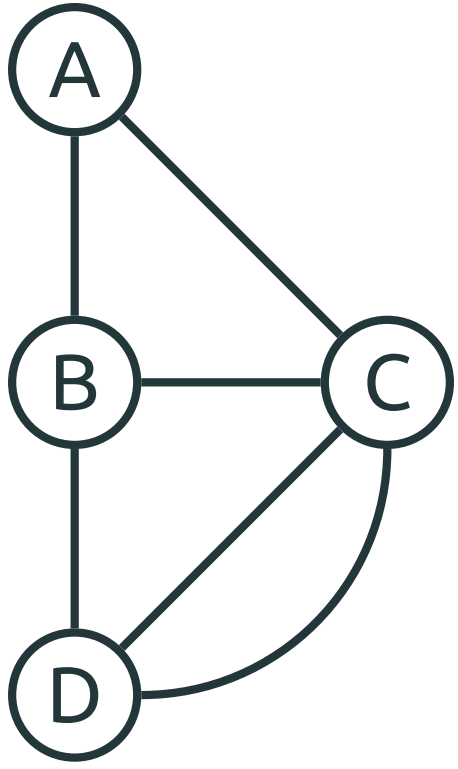
All but START and FINISH vertices
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# Six Bridges of Kaliningrad



# Six Bridges of Kaliningrad

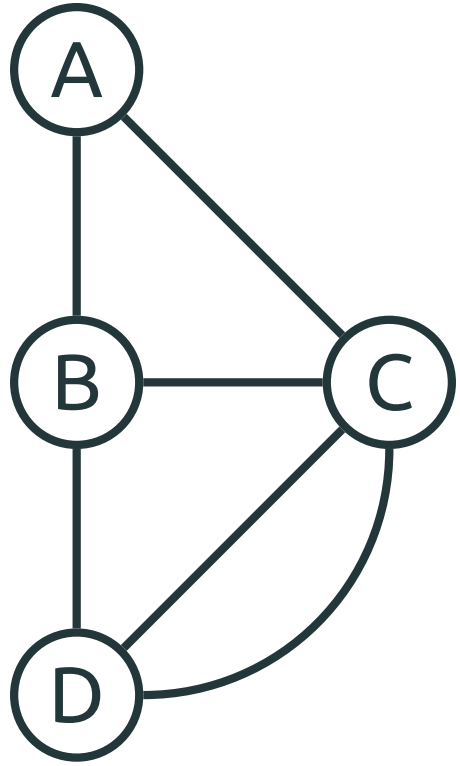




# Six Bridges of Kaliningrad

B and D have odd number of neighbors

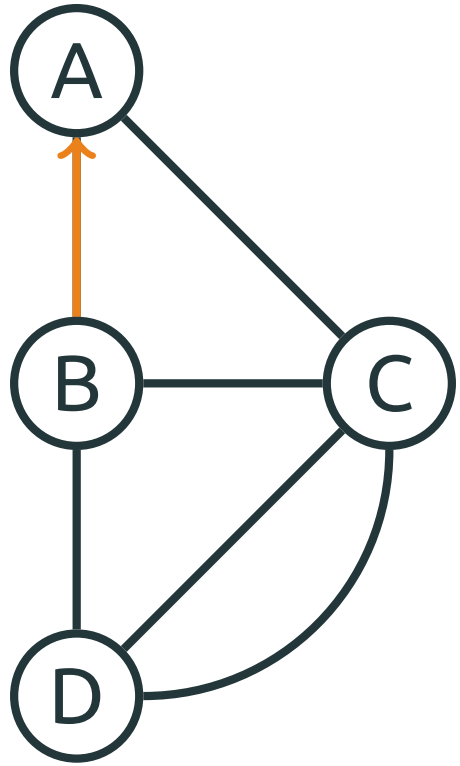
If there exists an Eulerian path, B and D must be START and FINISH



# Six Bridges of Kaliningrad

B and D have odd number of neighbors

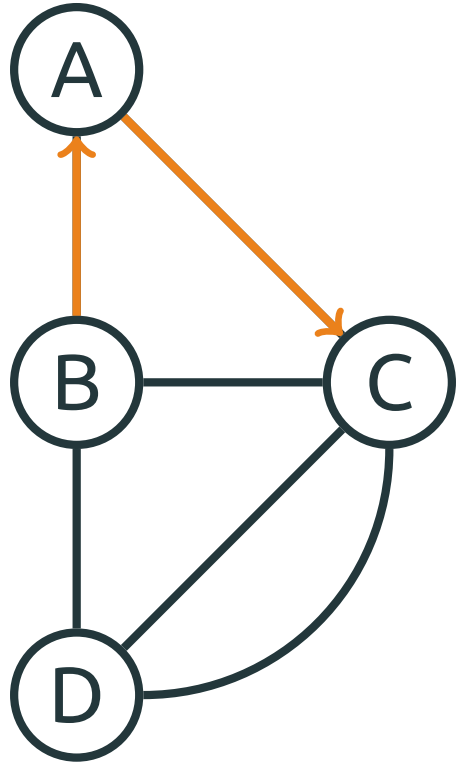
If there exists an Eulerian path, B and D must be START and FINISH



# Six Bridges of Kaliningrad

B and D have odd number of neighbors

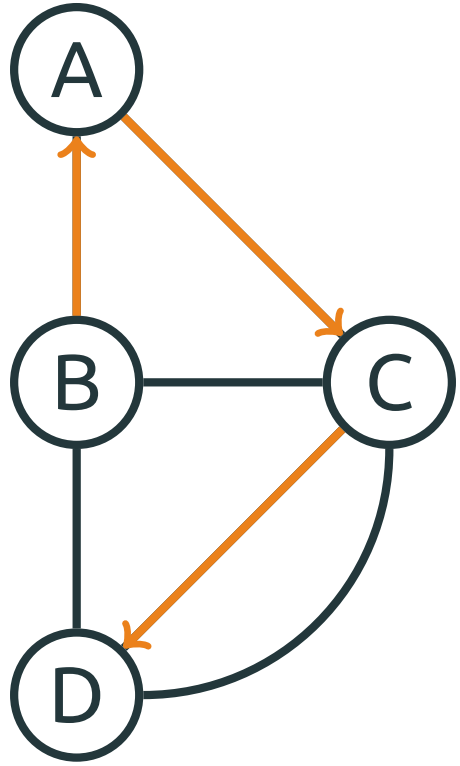
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# Six Bridges of Kaliningrad

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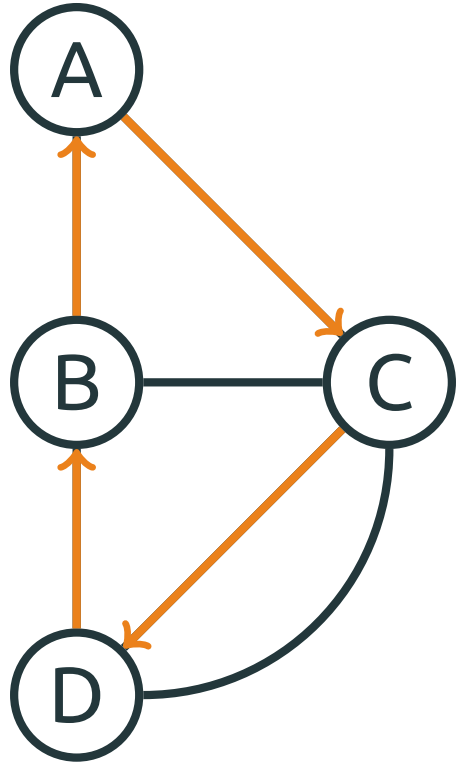
If there exists an Eulerian path, B and D must be START and FINISH



# Six Bridges of Kaliningrad

B and D have odd number of neighbors

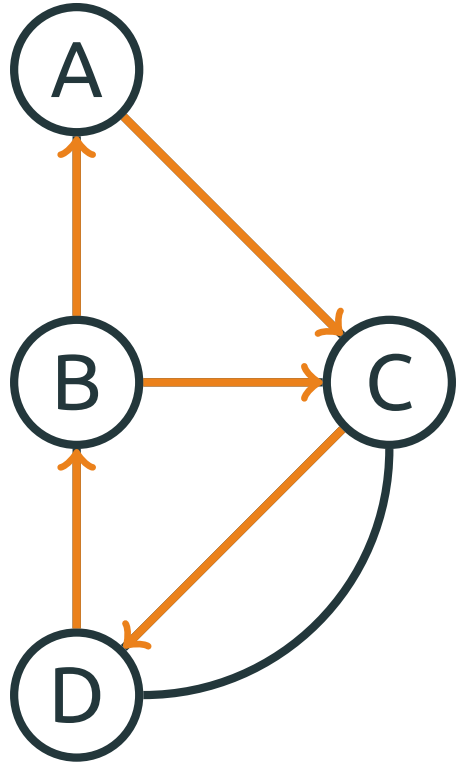
If there exists an Eulerian path, B and D must be START and FINISH



# Six Bridges of Kaliningrad

B and D have odd number of neighbors

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